Session 2: Using Agents

AI Agents for Descriptive Social-Science Research

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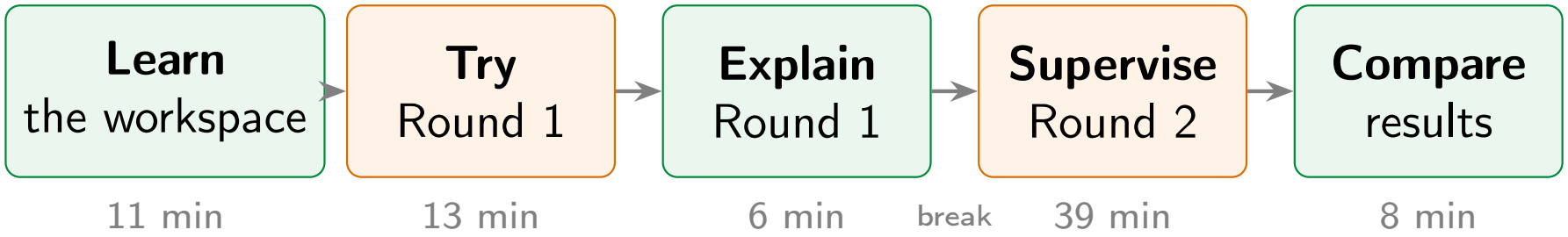
Before we start: open three things

Open these three things now, and keep them open all session. Give a **green tick** when you are ready.

1. In VS Code: **File ➜ Open Folder... ➜** `youth_at_home_project` (inside the course folder in Documents).
2. In the **Antigravity panel**, click + for a new conversation and type: *"What files are in this folder? Do not change anything."*
3. Open the **class form**: one Google Form where everyone enters their results.

➜ Something does not work? No problem. **Follow my screen** and do the supervisor's part: read the plan, decide, check.

Today: thirty supervisors, one figure



- **Learn:** the agent's workspace: folder, rules, locks, and the data.
- **Try** and **Supervise**: everyone runs the same task with their own agent.
- **Explain:** the words for what happened in Round 1.
- **Compare:** the class form collects your answers; we look at them together.

By the end, you can: set up an agent's workspace safely; see the decisions an agent takes for you; plan, approve, and check a figure from real data.

Part A

The agent's workspace

Structure is context

```
youth_at_home_project/
  README.md
  GEMINI.md
  data/
    raw/  (never touched)
    processed/
  scripts/
  outputs/
  docs/
  logs/
```

Black: in your folder now. Grey: a full project adds these.

- The agent learns about your project by looking at it. **Folder and file names are context.**
- `data/raw/` says "original data: do not change". `final_v2_NEW(3).csv` says nothing.
- A fixed structure means shorter prompts: "save it in `outputs/`" is enough.
- Grey folders: `processed/` for cleaned data, `docs/` for notes and codebooks, `logs/` for your **supervision log**: your record of working with the agent (more at the end of today).

House rules for any research folder

1. **Raw data are read-only.** Nothing writes in `data/raw/`. A backup copy lives elsewhere.
2. **Every result is made by a script.** No number is typed by hand or copied from the chat.
3. **One research question per folder.** A new question gets a new folder.
4. **Provenance is written down**, next to the data. In our folder: the *Data* section of `README.md`.

> **Provenance.** The documented origin of data: who produced them, where you got them, when, and what was done to them since.

Instruction file versus permission rules



Instruction file

Where

`GEMINI.md`, a file in your project folder

Written as

```
- Never edit, move, or delete
  files in data/raw/.
```

normal language

Checked by

the **LLM**

the model: it writes the text and decides what to do

What happens?

It **usually** obeys

It can forget, misread, or ignore it.

Instruction file versus permission rules

	Instruction file	Permission rules
Where	`GEMINI.md`, a file in your project folder	the tool's settings, *not* in `GEMINI.md`
Written as	`- Never edit, move, or delete files in data/raw/.` normal language	`edit data/raw/` **Deny** a menu, or a strict format
Checked by	the **LLM** the model: it writes the text and decides what to do	the **harness** the program around the LLM: it does the actions and checks the rules
What happens?	It **usually** obeys It can forget, misread, or ignore it.	**Blocked**, always Whatever the LLM writes.

Think of it as: a polite sign on the archive door, versus a locked door.

Instruction file versus permission rules



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What happens?	It **usually** obeys It can forget, misread, or ignore it.	**Blocked**, always Whatever the LLM writes.

Think of it as: a polite sign on the archive door, versus a locked door.

- ➜ Use **both**. Instructions explain what you want. Locks stop what must never happen.
- ➜ Today you edit only `GEMINI.md`. You meet the locks as **permission requests**: allow, ask, deny.

YOUR TURN Make the question broad (2 minutes)

For Round 1, we test what the agent does with a **broad** question. So we make the project files broad. Do it with me:

1. Open `GEMINI.md`. Replace the line that starts with `- Question:` with:

```
- Question: how common it is for young adults in the EU to live with their parents, and how this has changed.
```

2. Open `README.md`. Replace the line that starts with `**Research question.**` with:

```
**Research question.** How common is it for young adults in the EU to live with their parents, and how has this changed?
```

3. Save both files: `Ctrl+S` (Mac: `Cmd+S`).

Your line already says "how common"? Then your file is up to date. Just check it.

Allow, ask, or deny? You vote

Allow: runs without asking. **Ask**: stops for your permission each time. **Deny**: blocked, always.

Vote with Zoom reactions: **green tick** = allow, **raised hand** = ask, **red cross ("No")** = deny.

	The agent wants to…
1	read `README.md`
2	run `python scripts/plot.py`, which you approved
3	`pip install matplotlib` (add a package to Python)
4	change a file in `data/raw/`
5	download a table from Eurostat's website
6	delete the files in `outputs/` "to start clean"

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5	download a table from Eurostat's website	
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6	delete the files in `outputs/` "to start clean"	**deny**: ask why

Least privilege. Give the agent the minimum access it needs for this task, and nothing more. Start strict. Make a rule less strict only after you have seen the action many times.

Where the numbers come from

Eurostat. The statistical office of the European Union. It publishes comparable national data free online, in tables such as `ilc_lvps08`.

EU-SILC. European Union Statistics on Income and Living Conditions: a household survey run every year in each country. Our table comes from it.

API. A way for one program to ask another program for data, without anyone clicking.

Example: the script `scripts/01_download_eurostat.R` sends this web address. Eurostat sends back the whole table as a file.

`ec.europa.eu/eurostat/api/dissemination/sdmx/2.1/data/ilc_lvps08`

- Only the EU and some other countries. Your data describe only these.
- Public, aggregated statistics: safe to give the agent.

YOUR TURN

Round 1: one question, thirty agents

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10 minutes. Everyone asks the same question, word for word:

> In the EU, how much more likely are young men than young women to live with their parents? Answer in one sentence with one number. Do not create any files.

1. **Predict:** how many *different* numbers will thirty agents give? Type your guess in the Zoom chat. Press Enter only when I say "go".
2. In the Antigravity panel, click + (a new conversation). Normal mode, not plan mode. Paste the question exactly (it is also in the chat and in the form).
3. Read each permission request. Allow **this time only**.
4. Then ask: *"Which age group, which year, and which EU average did you use?"*
5. In the class form, choose **Round 1**, answer, and click **Submit**.

What did thirty agents say?

Same question, same data, same tool.

Why are the answers different?

Part B

Words that explain Round 1

Descriptive questions

Descriptive research question. A question about *what is the case*: how much, how many, where, when, and how it changed. It does not ask *why*.

Descriptive

- What share of young adults live with their parents in each country?
- Has that share risen since 2010?

Causal (not in this course)

- *Do* high rents *cause* young adults to stay home?
- Would cheaper housing *reduce* the share?

- **Our project question** (in `README.md`): how common is it for young adults in the EU to live with their parents, and how has this changed? Descriptive, but vague: *which* young adults, *which* EU, *which* years?
- Agents often explain causes ("because") too quickly. Our instruction file forbids it.

Observations and variables

Unit of observation. The kind of thing each row of the table describes. Here: **one group in one country in one year**, e.g. women aged 25–34 in Spain in 2025. Not one person.

Variable. A characteristic recorded for every observation, which can take different values. Here: age group (age), sex (sex), country (geo), year (`TIME_PERIOD`), the percentage (`OBS_VALUE`).

- The file has **nine age groups** (`Y16-19` ... `Y25-34`, `Y18-34`) and **23 years**.
- Your Round 1 prompt fixed the sex comparison, but not the age group or the year. **The agent chose them for you.**

Percentages and percentage points

> **Percentage point (pp).** The difference between two percentages, by simple subtraction. From 24.3% to 35.6% is **+11.3 percentage points**.

EU, ages 25–34, 2025: men **35.6%** live with their parents, women **24.3%**.

- Difference: $35.6 - 24.3 = $ **11.3 percentage points**.
- Ratio: $35.6/24.3 = 1.47$: men are **1.47 times as likely**, or "47% more likely".
- Same data, two correct numbers, very different messages. **Say which one you report.**

➜ Always ask: **percentage of what?** Here: men aged 25–34 living with their parents, divided by all men aged 25–34.

Gaps, warnings, and four EU averages

Missing value. An observation for which no value exists, e.g. a country that did not report a year. In our file a missing country-year simply has *no row*.

Flag. A letter Eurostat attaches to a value as a warning: **b** break in series (the method changed), **e** estimated, **p** provisional.

Four EU averages in the file:

- `EU27_2020`: today's 27 countries, **up to 2025**.
- `EU`, `EU28`, `EU27_2007`: older country groups that **stop in 2019**.

➜ An agent that picks `EU` gives you a 2019 number, and may call it "the latest".

Four answers, all reasonable

The agent's silent decisions	**Its answer**
`EU27_2020`, ages 25–34, 2025, difference	**11.3 pp** (35.6% vs 24.3%)
The same, as a ratio	**1.47 times** ("47% more likely")
Ages 18–34 instead	**9.1 pp** (54.3% vs 45.2%)
The `EU` average, which stops in 2019	**13.1 pp** (35.2% vs 22.1%)

- None of these is a hallucination. Each is a **decision the prompt left open**.
- Did your agent *tell* you which decisions it took, before you asked?

Eurostat ilc_lvps08, snapshot of 6 Nov 2026. Other combinations are possible.

Break

- If your Round 1 agent is still working, let it finish. Then submit Round 1 in the form.
- Keep VS Code and the project folder open.

YOUR TURN

Round 2: supervise a figure

YOUR TURN Write the decisions down (2 minutes)

1. Open `GEMINI.md` (click it in the Explorer).
2. Under `## Rules`, **delete** the line `- Write new files only in outputs/.`
3. In its place, type or paste these four lines:

```
- Save scripts in scripts/ and results in outputs/.
- "Young adults" means ages 25-34 (age code Y25-34). State every year.
- "EU average" means EU27_2020. Gaps and changes in percentage points.
- Keep Eurostat's flags (column OBS_FLAG) and list them in your report.
```

4. Save: `Ctrl+S` (Mac: `Cmd+S`).

➜ **You** change the rules, not the agent. That is the supervisor's job.

LIVE DEMO The Round 2 task, in plan mode

Same question as Round 1, now supervised. New conversation (+), then `/plan`, then this prompt:

> Read `README.md` and `GEMINI.md`. Then plan a Python script, saved in `scripts/`, that: (1) reads the raw Eurostat file; (2) keeps 2025, age `Y25-34`, men (`M`) and women (`F`), the 27 EU countries and `EU27_2020`; (3) computes the gap, men minus women, in percentage points; (4) saves a table with both values, their flags, and the gap to `outputs/`; (5) draws a dot plot to `outputs/`: one dot for men and one for women per country, countries sorted by the gap. After running it, describe the figure in three sentences. Report missing values and flags. **Plan only: do not create any files yet.**

LIVE DEMO Read the plan: five questions

For each question: **does the plan say it?** If the answer is *no* or *unclear*, correct it, or ask the agent.

1. **Question:** does the plan keep 2025, ages 25–34, and men minus women, in *percentage points*?
2. **Silent decisions:** does the plan *say* what it does with a missing 2025 value, and with flagged values (**b**, **e**, **p**)?
3. **Raw data:** does the plan leave `data/raw/` untouched, with nothing written there?
4. **Install, delete, send:** does the plan say which package it needs, and why? Does it delete or send anything?
5. **Checks:** does the plan say it will print the number of rows and the `EU27_2020` values, so you can check them?

YOUR TURN Round 2: same question, now supervised

Round 2: 20 minutes. Finished early? Then try the extra challenges.

1. **Predict** in the Zoom chat: in which EU country is the gap between young men and young women *largest*? Press Enter only when I say "go".
2. **Open the class form** and choose **Round 2**. The page has the prompt and the five questions.
3. **Plan**, then **read the plan** with the five questions. Correct or clarify at least one thing.
4. **Approve**. Allow each request **this time only**. A package install: ask why first.
5. **Submit**, post a **screenshot of your figure** in the Zoom chat, and give a **thumbs-up** reaction.

Everything you need is on the form's Round 2 page.
Work through it from top to bottom. Keep the tab open until you submit.

YOUR TURN Finished early? Two extra challenges

Submitted Round 2 first? Then choose one:

1. Over time	Add 2010 next to 2025. Has the gap between men and women grown or shrunk? Same file, one more year.
2. Your question	Your own descriptive question about this file: another age group, a line from 2003 to 2025, or one country in detail. Write the prompt with Tuesday's five parts: goal, input, output, constraints, check.

In every challenge, you stay the supervisor:

- ✓ Plan first. Check it with the five questions; correct or clarify at least one thing.
- ✓ Allow once, never "always". Install: ask why. Delete or move: refuse.
- ✓ Results only from scripts: no number typed by hand.
- ✗ The agent proposes something that is not in the plan? **Stop and ask why.**

Compare

What thirty supervisors found

Round 1 versus Round 2: the same question

Same question, same data, same agents.

Did the answers agree this time?

- What changed: **decisions written down**, and **you read the plan**.
- Which of the five questions did the first plans fail? Those are the decisions the agent still took alone.

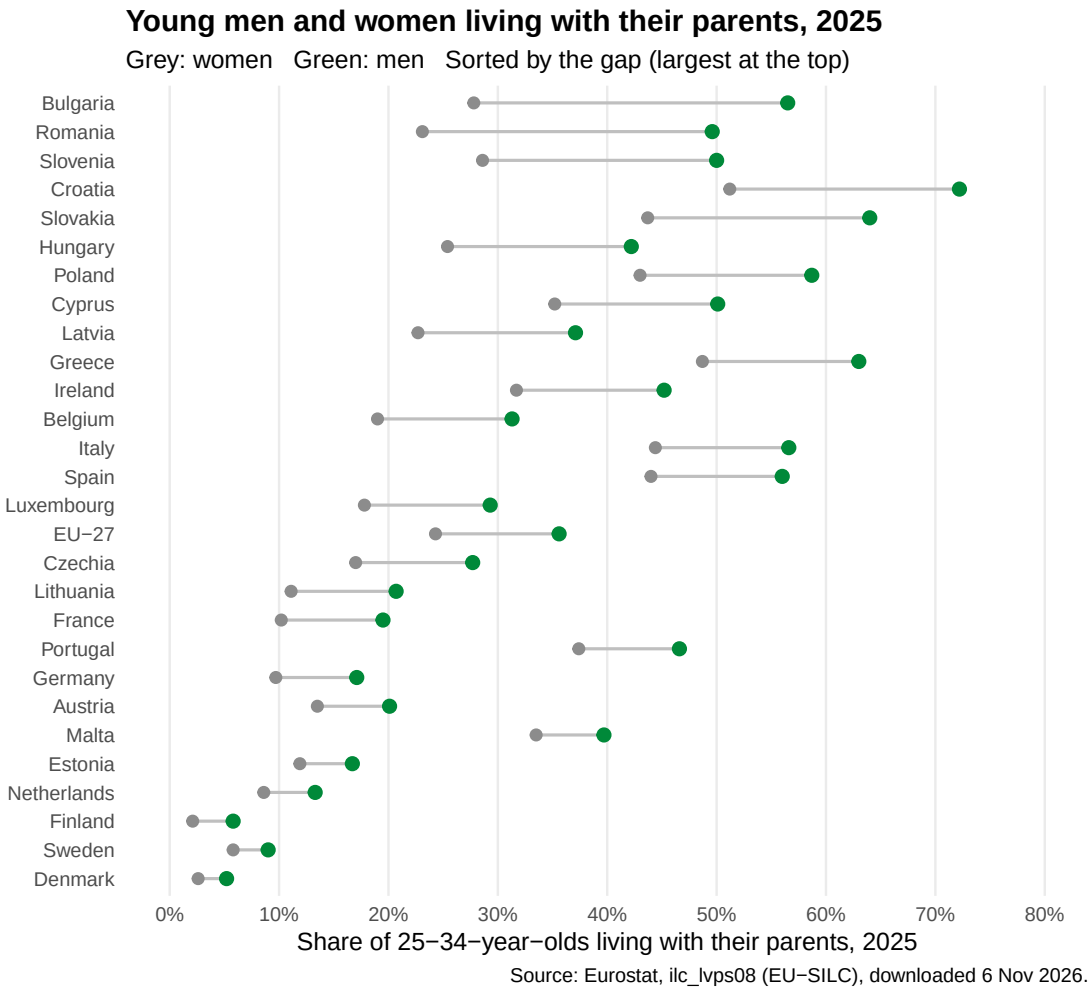
Where is the gap largest?

	Country	Men	Women	Gap		Country	Men	Women	Gap
1	Bulgaria	56.5	27.8	+28.7	15	Luxembourg	29.3 b	17.8 b	+11.5
2	Romania	49.6	23.1	+26.5		**EU-27**	**35.6**	**24.3**	**+11.3**
3	Slovenia	50.0	28.6	+21.4	16	Czechia	27.7	17.0	+10.7
4	Croatia	72.2	51.2	+21.0	17	Lithuania	20.7 p	11.1 p	+9.6
5	Slovakia	64.0	43.7	+20.3	18	France	19.5	10.2	+9.3
6	Hungary	42.2	25.4	+16.8	19	Portugal	46.6	37.4	+9.2
7	Poland	58.7	43.0	+15.7	20	Germany	17.1	9.7	+7.4
8	Cyprus	50.1	35.2	+14.9	21	Austria	20.1	13.5	+6.6
9	Latvia	37.1	22.7	+14.4	22	Malta	39.7	33.5	+6.2
10	Greece	63.0	48.7	+14.3	23	Estonia	16.7	11.9	+4.8
11	Ireland	45.2	31.7	+13.5	24	Netherlands	13.3	8.6	+4.7
12	Belgium	31.3	19.0	+12.3	25	Finland	5.8	2.1	+3.7
13	Italy	56.6	44.4	+12.2	26	Sweden	9.0	5.8	+3.2
14	Spain	56.0	44.0	+12.0	27	Denmark	5.2	2.6	+2.6

Young men are more likely than young women to live with their parents in **all 27 countries**.

Eurostat ilc_lvps08, 2025, ages 25–34, % living with their parents; gap = men minus women, in percentage points; snapshot of 6 Nov 2026. Flags: b break in series, p provisional.

The result (one possible version)



It says

- Men are above women in **all 27 countries**.
- Largest gap: **Bulgaria** +28.7 pp. Smallest: **Denmark** +2.6 pp.
- **EU-27**: 35.6% vs 24.3%, **+11.3 pp**.

It does not say

- *Why*: jobs, studies, partners, housing, norms, war. . .
- Anything about individuals, regions, or Ukraine.

Limitations of our figure

1. **Only EU countries.** No Ukraine, and no regions inside countries.
2. **Two warnings in the data.** Luxembourg 2025: the method changed (**b**). Lithuania 2025: provisional (**p**).
3. **Who is counted.** The survey asks private households only. Young people in student halls, barracks, or care homes are not in it.
4. **Survey numbers.** They come from a sample, so a small gap (Denmark: 2.6 points) may be chance.
5. **One year only.** 2025 does not show if the gap is growing or shrinking.

➜ **Write the limitations next to the figure.** Then ask your agent for more limitations and improvements, and check its answer. 😎

If we did it again: five levels of supervision

5. Check, log, rerun.
Next time

4. Read and correct the plan.
Round 2

3. Write decisions down first.
After the break, in `GEMINI.md`

2. Ask for its choices.
Round 1, follow-up question

1. Take the number.
Round 1

Today in four sentences

1. A **standard folder** is context: raw data untouched, every result made by a script.
2. Instructions *ask*; **settings lock**. Allow, ask, deny.
3. **Know your data**: what each row is, the flags, and what is missing.
4. A question you leave open is **decided silently**, so decide it yourself.

Your week: go big 😎

Your own descriptive question.

Instructions: tonight, by email and in the course folder on GitHub:

`github.com/Martinalberdi/ai_agents_social_science`

On Tuesday: we look at what you built, and at what your agent assumed.

Questions?